

## Lesson 4.1 - Introduction to Polynomials

Learning Objectives: SWBAT

1. Classify Polynomials by degree and number of terms
2. Evaluate Polynomials
3. Write a polynomial in standard form

Making a connection

- In unit 3, we learned about QUADRATIC functions/equations. We know that all quadratics have the following in common:
  - > They are "\_\_\_\_\_ Degree" equations
  - > They have a general shape of the letter \_\_\_\_\_ with exactly \_\_\_\_\_ max or min
  - > They have "solutions" (values of  $x$  that result in a  $y$  that equals \_\_\_\_\_)
  - > They all have a domain of \_\_\_\_\_
  - > They all have an axis of \_\_\_\_\_
- A quadratic is a specific kind of POLYNOMIAL. To understand polynomials better, let's take a look at some important vocabulary that we will use throughout the unit:

A **monomial** is a numeral, a variable, or the product of a numeral and one or more variables. A monomial with no variables, such as  $-1$  or  $\frac{2}{3}$ , is called a **constant**. (in essence, a monomial is an expression with one "term")

A **coefficient** is the numerical factor in a monomial. For example:

- $x$  has a coefficient of 1.
- $-2t$  has a coefficient of  $-2$ .
- $\frac{-x^3y^2}{3}$  has a coefficient of  $-\frac{1}{3}$  because it can be written as  $-\frac{1}{3}x^3y^2$ .
- $-ab$  has a coefficient of  $-1$ .

A **polynomial** is a monomial or a sum of terms that are monomials. In this chapter, you will study polynomials in one variable. Polynomials can be classified by the number of terms they contain. A polynomial with two terms is a **binomial**. A polynomial with three terms is a **trinomial**.

The **degree of a polynomial** is the same as that of its term with the greatest degree. Polynomials can also be classified by degree, as shown below.

| CLASSIFICATION OF A POLYNOMIAL BY DEGREE |           |                                      |
|------------------------------------------|-----------|--------------------------------------|
| Degree                                   | Name      | Example                              |
| $n = 0$                                  | constant  | 3                                    |
| $n = 1$                                  | linear    | $5x + 4$                             |
| $n = 2$                                  | quadratic | $-x^2 + 11x - 5$                     |
| $n = 3$                                  | cubic     | $4x^3 - x^2 + 2x - 3$                |
| $n = 4$                                  | quartic   | $9x^4 + 3x^3 + 4x^2 - x + 1$         |
| $n = 5$                                  | quintic   | $-2x^5 + 3x^4 - x^3 + 3x^2 - 2x + 6$ |

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### Classifying Polynomials (Examples)

Classify each polynomial by degree and by number of terms.

a.  $2x^3 - 3x + 4x^5$

b.  $-2x^3 + 3x^4 + 2x^3 + 5$

#### SOLUTION

a. The greatest exponent of  $x$  is 5, so the degree is 5.

The polynomial has three terms, so it is a trinomial.

The polynomial is a quintic trinomial.

b. When simplified to  $3x^4 + 5$ , the greatest exponent of  $x$  is 4, so the degree is 4.

The simplified polynomial has two terms, so it is a binomial.

The polynomial is a quartic binomial.

#### Your Turn

Classify each polynomial by degree and by number of terms.

a.  $x^2 + 4 - 8x - 2x^3$

b.  $3x^3 + 2 - x^3 - 6x^5$

### Evaluating Polynomials

- Polynomials are "evaluated" the same way that any other function is evaluated.
- Simply plug in the number that you are evaluating in for the variable and solve

> Example: Evaluate  $f(x) = 3x^4 + 4x^3 - 5x^2 + 6x - 2$  for  $x = 2$

> Solution:  $f(2) = 3(2)^4 + 4(2)^3 - 5(2)^2 + 6(2) - 2 = \boxed{70}$

- Your Turn: Evaluate  $f(x) = 2x^4 + x^3 - 3x^2 + 5x$ ,  $x = -1$

### Standard Form

- Simply put, standard form of any polynomial equation comes down to "Descending order of Degree" - The highest degree term is furthest left and each subsequent term is one degree less than the previous term. The furthest term on the right is always a constant

> For example  $4x^6 + 3x^5 + 8x^4 + 7x^3 - x^2 + 5x^1 + 13x^0$

- Remember, each degree must be accounted for. If you see a polynomial with any "missing" degrees, we can complete the sequence using "invisible" zeros

> For example  $4x^6 + 8x^4 - x^2 + 13$  can be re-written as:

$$4x^6 + 0x^5 + 8x^4 + 0x^3 - x^2 + 0x^1 + 13$$

## **Lesson 4.1 - Introduction to Polynomials**

Practice:

**Name each polynomial by degree and number of terms.**

1)  $2p^4 + p^3$

2)  $-10a$

3)  $2x^2$

4)  $-10k^2 + 7$

5)  $-5n^4 + 10n - 10$

6)  $-6a^4 + 10a^3$

7)  $6n$

8)  $1$

9)  $-9n + 10$

10)  $5a^2 - 6a$

11)  $8p^5 - 5p^3 + 2p^2 - 7$

12)  $-7n^7 + 7n^4$

13)  $-8n^4 + 5n^3 - 2n^2 - 8n$

14)  $9v^7 + 7v^6 + 4v^3 - 1$

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Practice:

Answer the questions or identify the specified parts of the polynomial:

$$5x^3 - 2x^2 + 14x + 7x^2 + 3x - 11$$

- a. How many terms does this polynomial have? \_\_\_\_\_
- b. Write the term that has a degree of 3. \_\_\_\_\_
- c. What is the coefficient of this term with degree 3? \_\_\_\_\_
- d. Which term is a constant term? \_\_\_\_\_
- e. List a pair of like terms. \_\_\_\_\_ and \_\_\_\_\_
- f. List another pair of like terms. \_\_\_\_\_ and \_\_\_\_\_
- g. Rewrite the polynomial in simplified form by combining the like terms:
  
- h. Evaluate this polynomial for  $x = 2$ . \_\_\_\_\_

**Evaluate each function at the given value.**

1)  $f(x) = 3x^3 + 8x^2 + 7x + 1$  at  $x = -2$

2)  $f(n) = 3n^2 + 7n - 10$  at  $n = -3$

3)  $f(a) = 2a^3 - 3a^2 - 3a$  at  $a = 2$

4)  $f(n) = -n^2 + 9$  at  $n = -3$

5)  $f(n) = n^3 - 5n^2 + 9n - 13$  at  $n = 3$

6)  $f(n) = n^2 - 6n + 7$  at  $n = 3$

7)  $f(n) = n^4 + 10n^3 + 22n^2 - 15n - 22$  at  $n = -4$